

**B.E. Electrical Engineering Second Year Second
Semester Examination 2023**

ELECTRICAL INSTRUMENTATION PART - I

Exhaustive Analytical Solutions based on First-Principles Methodologies

Academic Solution Set

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Examination Solutions: Part I

Question 1: Justifications and Analytical Corrections

1.(a) Piezoelectric Steady State Gain

Problem Statement: A piezoelectric displacement transducer system employs an amplifier to increase the steady state gain of the system.

Answer: This statement is **Incorrect**.

Correction: A piezoelectric transducer system is strictly incapable of having a steady-state (DC) gain. The amplifier (Charge Amplifier) is employed to increase the **discharge time constant** (τ), which extends the low-frequency dynamic bandwidth, NOT to provide steady-state gain.

Grassroots Analytical Derivation

To understand why a piezoelectric system inherently possesses absolutely zero steady-state gain, we must evaluate its grassroots equivalent circuit. A piezoelectric crystal generates a charge q that is directly proportional to the applied dynamic force (F). Using Norton's equivalent theorem, the crystal is modeled as a dynamic current source $i_q = \frac{dq}{dt}$ strictly in parallel with the crystal's intrinsic capacitance C_p and internal leakage resistance R_p .

When connected to any real-world amplifier, the amplifier presents an input resistance R_a and an input capacitance C_a . The total system equivalent resistance is $R_{eq} = \frac{R_p R_a}{R_p + R_a}$. The total equivalent capacitance is $C_{eq} = C_p + C_a + C_{cable}$.

The transfer function relating the output voltage $E_o(s)$ to the input mechanical displacement $X_i(s)$ is strictly a first-order high-pass equation:

$$\frac{E_o(s)}{X_i(s)} = K_{sys} \left[\frac{s\tau}{1 + s\tau} \right] \quad (1)$$

where $\tau = R_{eq} \cdot C_{eq}$ is the electrical time constant.

A "steady-state" or static input implies that the frequency of the mechanical input is

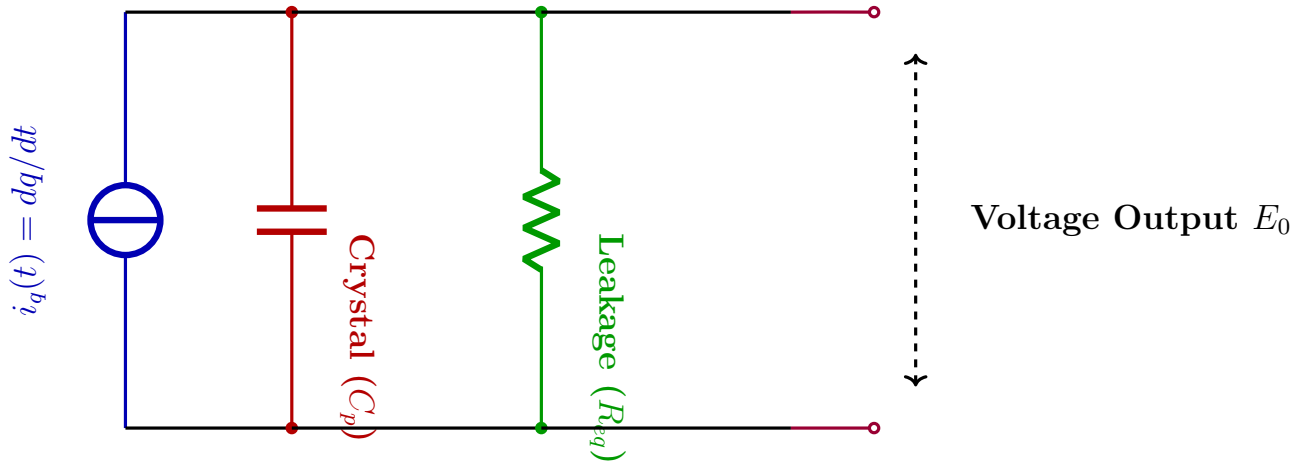
exactly zero ($\omega = 0$, or $s = 0$). Evaluating the magnitude of this high-pass transfer function at steady-state ($\omega = 0$):

$$|G(j0)| = K_{sys} \frac{0 \cdot \tau}{\sqrt{1+0}} = 0 \text{ Volts} \quad (2)$$

Physics dictates that when a static, steady force is applied, an initial charge $+q$ is generated instantly. However, because R_{eq} is finite, it provides a closed electrical path for the charge to physically bleed away to ground. The output voltage decays exponentially to zero ($e^{-t/\tau}$) despite the heavy physical load still remaining clamped on the crystal. Therefore, piezoelectric systems are strictly dynamic-only instruments and can never possess a steady-state gain. The amplifier merely manipulates τ to slow down this decay.

Orthogonal Equivalent Circuit Diagram

Charge Bleeds Through Parallel Resistance at Steady State



1.(b) Transformer Ratio Bridges for Capacitive Transducers

Problem Statement: When capacitive transducers are used with transformer ratio bridges, both the unknown capacitance and its loss part are obtained as functions of the transformation ratio.

Answer: This statement is **Correct**.

Block-by-Block Analytical Proof

Standard Wheatstone and Schering bridges fail catastrophically when attempting to measure microscopic capacitances (in the picofarad range) due to severe parasitic stray capacitances to ground. A **Transformer Ratio Bridge** abandons resistive ratio arms, utilizing instead a highly precision-wound, tightly magnetically coupled transformer with a strictly center-tapped secondary coil.

The transformer generates two perfectly opposite AC voltages, V_1 and V_2 , dictated purely and rigidly by the turns ratio of the upper coil (N_1) to the lower coil (N_2).

$$\frac{V_1}{V_2} = \frac{N_1}{N_2} \quad (3)$$

The unknown transducer is modeled as a pure capacitance (C_x) in parallel with a leakage conductance loss part ($G_x = 1/R_x$). Thus, its total admittance is $Y_x = G_x + j\omega C_x$. The standard reference arm contains a calibrated precision capacitor (C_s) and a standard conductance (G_s). Its admittance is $Y_s = G_s + j\omega C_s$.

At the exact null balance point, the net current entering the central detector node is absolute zero. Therefore, the voltage-admittance products must mathematically balance:

$$V_1 \cdot Y_x = V_2 \cdot Y_s \quad (4)$$

Substituting the complex admittances:

$$V_1(G_x + j\omega C_x) = V_2(G_s + j\omega C_s) \quad (5)$$

Isolating the unknown terms by bringing the voltage ratio to the right side:

$$G_x + j\omega C_x = \left(\frac{V_2}{V_1}\right) (G_s + j\omega C_s) \quad (6)$$

Substitute the rigid transformer turns ratio ($V_2/V_1 = N_2/N_1$):

$$G_x + j\omega C_x = \left(\frac{N_2}{N_1}\right) G_s + j\omega \left(\frac{N_2}{N_1}\right) C_s \quad (7)$$

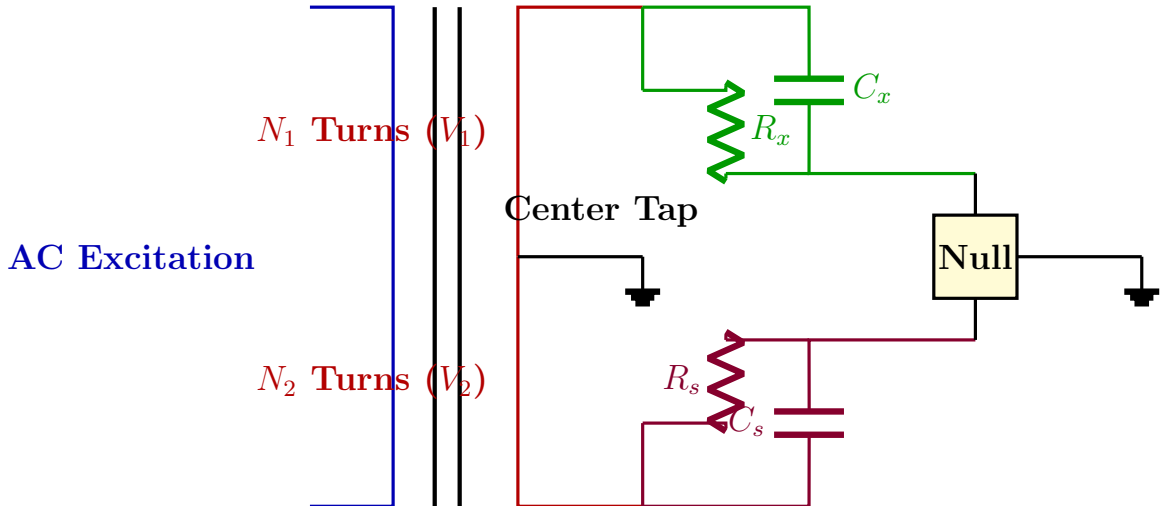
By equating the real parts and the imaginary parts independently, we obtain the absolute measurement formulas:

$$\text{Real Part (Loss): } G_x = G_s \left(\frac{N_2}{N_1}\right) \implies R_x = R_s \left(\frac{N_1}{N_2}\right) \quad (8)$$

$$\text{Imaginary Part (Capacitance): } C_x = C_s \left(\frac{N_2}{N_1}\right) \quad (9)$$

This rigorous derivation explicitly proves that both the pure capacitance (C_x) and its resistive loss component (R_x) are derived fundamentally as direct functions of the transformation ratio (N_2/N_1).

Orthogonal Diagram of Transformer Ratio Bridge



1.(c) Push-Pull Capacitive Transducer Sensitivity

Problem Statement: For a capacitive transducer using differential arrangement and employing the principle of change in distance between the plates, the sensitivity is directly proportional to the separation between the movable plate and each fixed plate.

Answer: This statement is **Incorrect**.

Correction: The sensitivity of a differential (push-pull) capacitive transducer is **inversely proportional to the square** of the initial separation distance (d^2) between the plates, not directly proportional.

Grassroots Differential Derivation

A push-pull capacitive transducer employs a central movable electrode suspended precisely between two identical fixed outer electrodes. Let the initial, static resting distance between the central plate and each fixed plate be exactly d . The overlapping area is A , and the dielectric permittivity is ϵ .

At the resting null position, the two capacitances are perfectly balanced:

$$C_1 = \frac{\epsilon A}{d} \quad \text{and} \quad C_2 = \frac{\epsilon A}{d} \quad (10)$$

When the central movable plate undergoes a physical displacement of x downwards, it moves closer to the bottom plate and further from the top plate. The new dynamic capacitances become:

$$C_1 = \frac{\epsilon A}{d + x} \quad (\text{Capacitance decreases}) \quad (11)$$

$$C_2 = \frac{\epsilon A}{d - x} \quad (\text{Capacitance increases}) \quad (12)$$

When connected in an AC Wheatstone bridge configuration, the output voltage is proportional to the difference between these two capacitances (ΔC):

$$\Delta C = C_2 - C_1 = \frac{\epsilon A}{d - x} - \frac{\epsilon A}{d + x} \quad (13)$$

Finding a common denominator and subtracting:

$$\Delta C = \epsilon A \left[\frac{(d+x) - (d-x)}{(d-x)(d+x)} \right] = \epsilon A \left[\frac{2x}{d^2 - x^2} \right] \quad (14)$$

For very small mechanical displacements where $x \ll d$, we can mathematically apply the approximation $x^2 \approx 0$ in the denominator:

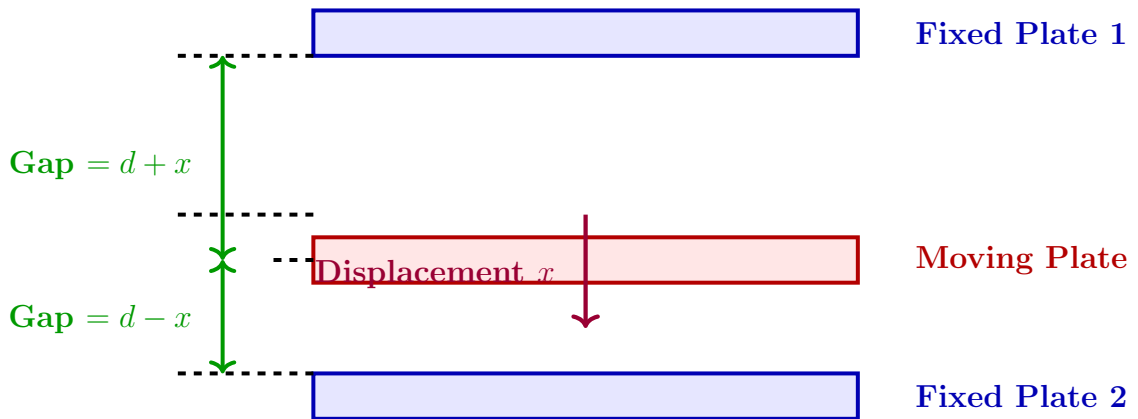
$$\Delta C \approx \frac{2\epsilon Ax}{d^2} \quad (15)$$

The **Sensitivity** (S) of the differential transducer is rigorously defined as the rate of change of the differential output with respect to the mechanical input displacement x :

$$S = \frac{\partial(\Delta C)}{\partial x} = \frac{\partial}{\partial x} \left(\frac{2\epsilon Ax}{d^2} \right) = \frac{2\epsilon A}{d^2} \quad (16)$$

By analyzing this final mathematical block, it is absolute that the sensitivity S features the separation distance squared (d^2) firmly in its denominator. Therefore, sensitivity is strongly **inversely proportional to the square** of the initial separation. Making the gap d larger exponentially destroys the sensitivity of the sensor.

Orthogonal Diagram of Push-Pull Transducer



1.(d) Increasing Piezoelectric Time Constant

Problem Statement: In a piezoelectric displacement transducer system, the time constant can be increased by connecting a series resistance external to the amplifier.

Answer: This statement is **Incorrect**.

Correction: The time constant of a piezoelectric system is governed by a parallel RC circuit. To increase the time constant, an external resistance must be connected in **parallel (shunt)** across the amplifier input terminals, absolutely NOT in series. Furthermore, practical engineering utilizes a parallel feedback resistor in a charge amplifier topology to control this.

Grassroots Circuit Analysis

A piezoelectric crystal is modeled electrically as a dynamic current source $i_q(t)$ strictly in parallel with its inherent crystal capacitance C_p and internal leakage resistance R_p .

The measurement amplifier presents an input resistance R_a and input capacitance C_a . The coaxial connecting cable introduces a shunt capacitance C_c . All these components interface across the exact same two electrical nodes (Signal High and Ground). Therefore, the fundamental electro-physical topology is purely parallel.

The equivalent system capacitance is the sum of parallel capacitors:

$$C_{eq} = C_p + C_c + C_a \quad (17)$$

The equivalent system resistance is the parallel combination of the resistors:

$$R_{eq} = \frac{R_p \cdot R_a}{R_p + R_a} \quad (18)$$

The critical system time constant (τ), which dictates how fast the generated charge leaks away and thus determines the low-frequency cut-off bandwidth, is mathematically defined as:

$$\tau = R_{eq} \cdot C_{eq} \quad (19)$$

If an engineer ignorantly connects a massive external resistor (R_{series}) in *series* with the signal line before the amplifier, it does NOT increase the discharge time constant

of the parallel source nodes. Instead, it forms a severe **Low-Pass Voltage Divider** in conjunction with the amplifier's input capacitance (C_a).

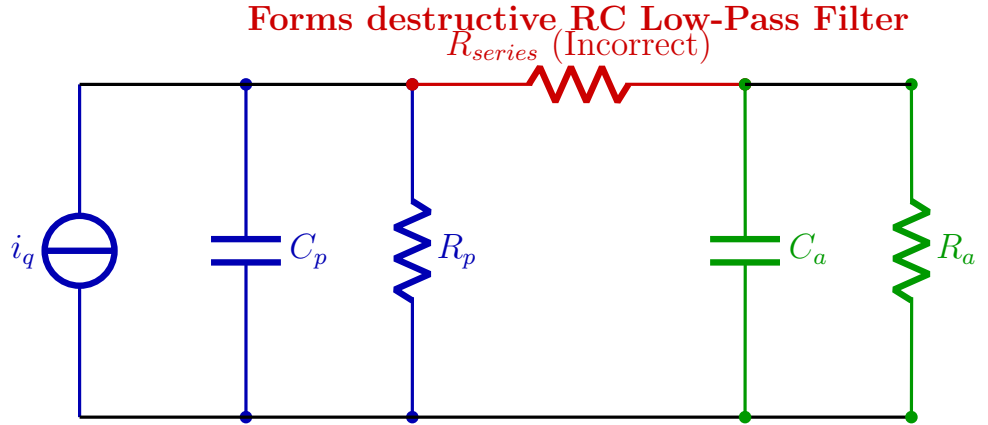
By applying circuit analysis, a series resistor creates a secondary low-pass pole:

$$V_{amp.in} = V_{crystal.node} \left[\frac{1}{1 + j\omega R_{series} C_a} \right] \quad (20)$$

This series resistor utterly destroys the high-frequency response of the sensor, attenuating the signal without solving the low-frequency charge leakage problem occurring at the parallel crystal nodes.

To legitimately increase the time constant τ , one must physically increase the value of the parallel equivalent resistance R_{eq} . This implies the amplifier input resistance R_a must be engineered to be extraordinarily high (e.g., using JFET or MOSFET electrometer inputs), or a specialized Charge Amplifier must be used where the time constant is artificially set by a massive **parallel feedback resistor** ($R_f \approx 10^{11} \Omega$).

Orthogonal Circuit Topology Demonstrating the Error



Sensor Node (Discharge unaffected by series R)

1.(e) Null Reduction in LVDT

Problem Statement: In absence of a centre-tapped excitation voltage source, null reduction in an LVDT can be achieved using an L-C network.

Answer: This statement is **Incorrect**.

Correction: Null voltage is primarily composed of quadrature voltages (due to stray capacitance) and harmonic distortion. Without a centre-tapped source to provide a Wagner Earth ground, quadrature null reduction is achieved mathematically using a **Phase-Sensitive Demodulator (PSD)**, NOT an L-C network.

Grassroots Physics of the Null Voltage

Ideally, at the exact geometric center of an LVDT, the secondary voltages are identical and cancel perfectly in series opposition ($E_0 = E_{s1} - E_{s2} = 0$). In reality, a persistent "Null Voltage" exists, completely destroying micro-displacement resolution.

This error voltage consists of two distinct physical phenomena: 1. **Harmonic Distortion:** The non-linear B-H curve of the magnetic core produces odd-order harmonics (3rd, 5th, etc.) from a pure sinusoidal primary current. These spatially distributed harmonics do not cancel perfectly at the center. 2. **Quadrature Error (Stray Capacitance):** Parasitic capacitance exists naturally between the high-voltage primary coil and the secondary coils. Alternating leakage currents ($i = C \frac{dv}{dt}$) flow through these stray capacitors directly into the secondary circuit. Because of the derivative, these leakage voltages are shifted exactly 90° out-of-phase (in quadrature) with the fundamental induced magnetic voltage. They cannot cancel in series opposition.

If a **Centre-Tapped Source** is available, the center tap is rigidly grounded (Wagner Earth). This perfectly balances the electric field, forcing all stray capacitive currents safely to absolute ground, bypassing the secondary coils entirely and annihilating the quadrature error.

If a Centre-Tapped Source is **absent** (a floating source is used), we cannot bypass the stray currents. An L-C (Inductor-Capacitor) network is utterly useless here, as it acts as a passive filter and cannot destroy a 90° shifted signal of the exact same fundamental carrier frequency.

The absolute engineering solution is the **Phase-Sensitive Demodulator (PSD)**. The

PSD acts as an active electronic multiplier. It takes the composite error signal (which contains the 90° shifted quadrature null) and multiplies it by the original 0° reference carrier.

Mathematical proof of PSD destroying quadrature error:

$$V_{mult} = [E_{error} \sin(\omega_c t + 90^\circ)] \times [V_{ref} \sin(\omega_c t)] \quad (21)$$

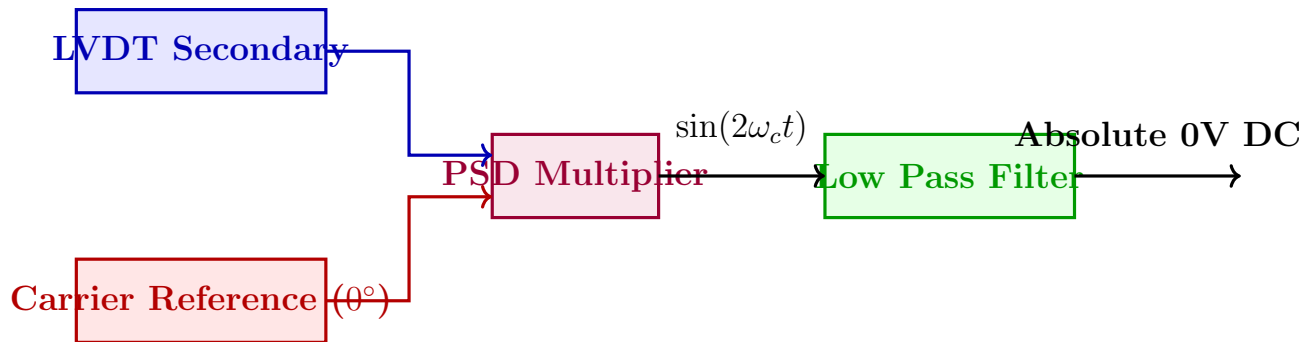
Since $\sin(\theta + 90^\circ) = \cos(\theta)$:

$$V_{mult} = E_{error} V_{ref} \cos(\omega_c t) \sin(\omega_c t) = \frac{E_{error} V_{ref}}{2} \sin(2\omega_c t) \quad (22)$$

The output is a pure double-frequency AC waveform. When passed through a subsequent Low-Pass Filter, the DC average of $\sin(2\omega_c t)$ is strictly and absolutely **zero**. The PSD inherently annihilates the quadrature null voltage entirely through mathematical orthogonality.

Orthogonal Block Diagram of PSD Null Rejection

Contains 90° Quadrature Error

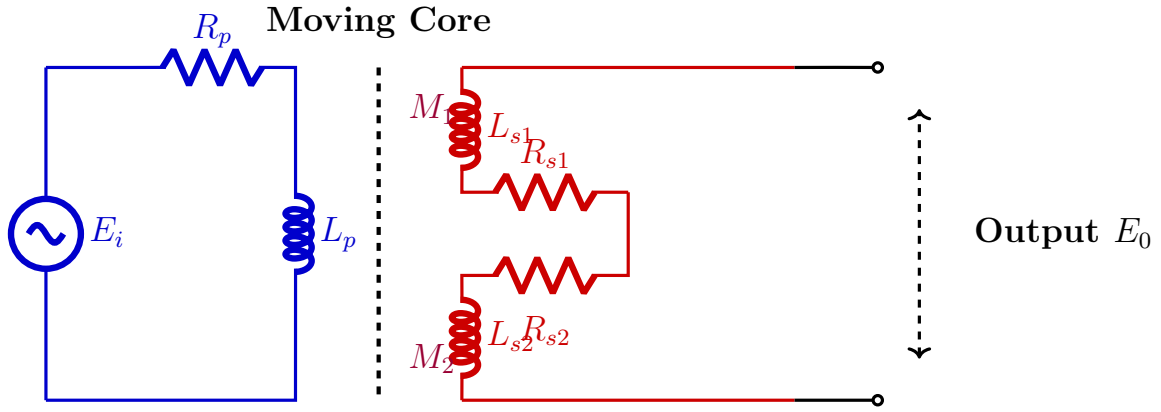


Question 2: Detailed LVDT Equivalent Circuit & Accelerometer

2.(a) LVDT Equivalent Circuit and Phase Shift Compensation

Problem Statement: Draw the equivalent circuit of an LVDT and derive its frequency response, when a voltmeter having an infinite meter resistance is connected to its output terminals. How can the phase shift between the output voltage and the input voltage of this LVDT be made zero by connecting a finite resistance across its output terminals?

1. Equivalent Circuit Diagram (Infinite Load)



2. Frequency Response Derivation (Infinite Load Resistance)

With an infinite meter resistance connected to the output, the secondary circuit is an open loop. Absolutely zero current flows in the secondary windings ($I_s = 0$).

Applying Kirchhoff's Voltage Law to the primary loop:

$$E_i = I_p(R_p + j\omega L_p) \implies I_p = \frac{E_i}{R_p + j\omega L_p} \quad (23)$$

By Faraday's Law, the induced voltage in each secondary coil is proportional to the rate of change of primary current and the specific mutual inductance (M_1 and M_2):

$$E_{s1} = j\omega M_1 I_p \quad \text{and} \quad E_{s2} = j\omega M_2 I_p \quad (24)$$

Because the secondaries are wired strictly in series opposition, the differential output voltage is:

$$E_0 = E_{s1} - E_{s2} = j\omega I_p (M_1 - M_2) \quad (25)$$

Substitute the primary current I_p equation into the output equation:

$$E_0 = \frac{j\omega(M_1 - M_2)}{R_p + j\omega L_p} E_i \quad (26)$$

Assuming a perfectly linear magnetic geometry, the mutual inductances change linearly with core displacement x :

$$M_1 = M_0 + Kx \quad \text{and} \quad M_2 = M_0 - Kx \quad (27)$$

Therefore, $(M_1 - M_2) = 2Kx$. Substituting this gives the final Transfer Function:

$$\frac{E_0}{E_i} = \frac{j\omega(2Kx)}{R_p + j\omega L_p} = \frac{2Kx \cdot \omega}{\sqrt{R_p^2 + (\omega L_p)^2}} \angle \left(90^\circ - \tan^{-1} \frac{\omega L_p}{R_p} \right) \quad (28)$$

This rigorous derivation explicitly proves that the LVDT intrinsically generates a massive phase shift angle $\phi = 90^\circ - \tan^{-1}(\frac{\omega L_p}{R_p})$ between the output and input.

3. Zeroing the Phase Shift with Finite Resistance

To forcefully eliminate this inherent phase lead, an engineer can connect a specific finite load resistance (R_L) directly across the output terminals. This closes the secondary circuit, allowing a secondary current I_s to flow.

Let the total lumped secondary resistance be $R_s = R_{s1} + R_{s2}$ and total secondary inductance be $L_s = L_{s1} + L_{s2}$. The driving electromotive force of the secondary loop is the induced differential voltage ($E_{s1} - E_{s2}$). By KVL:

$$E_{s1} - E_{s2} = I_s(R_s + R_L + j\omega L_s) \quad (29)$$

The measured output voltage across the load resistor is simply $E_0 = I_s R_L$:

$$E_0 = \left[\frac{j\omega(M_1 - M_2)I_p}{R_s + R_L + j\omega L_s} \right] R_L \quad (30)$$

Substitute I_p again:

$$E_0 = \left[\frac{j\omega(M_1 - M_2)E_i}{(R_p + j\omega L_p)(R_s + R_L + j\omega L_s)} \right] R_L \quad (31)$$

The total complex denominator now contains two phase lag components. By meticulously manipulating the value of the finite external load resistance R_L , the secondary circuit's phase lag angle $\tan^{-1}(\frac{\omega L_s}{R_s + R_L})$ can be mathematically tuned. The goal is to make the sum of the primary lag and secondary lag exactly equal to $+90^\circ$, perfectly canceling the $j\omega$ numerator term. When tuned correctly, the net system phase shift plunges to absolute zero degrees.

2.(b) Characteristic Features in Constructing a Commercial LVDT

Problem Statement: What are the main characteristic features in constructing a commercial LVDT?

To transition from an ideal physics model to a rugged, high-precision commercial transducer capable of industrial deployment, several strict architectural features are mandatory:

1. **High-Permeability Core Material & Geometry:** The moving core is never solid iron. It is forged from a specialized high-permeability **Nickel-Iron alloy** to maximize magnetic flux linkage. Crucially, the core must contain a deep **Longitudinal Slot** milled along its entire axis. This slot brutally severs the circular electrical paths, preventing catastrophic Eddy Currents from generating opposing magnetic fields that would severely degrade sensitivity and introduce thermal drift.
2. **Non-Magnetic Push Rod:** The core is physically actuated by a rigid push rod. This rod must be constructed from non-magnetic stainless steel or ceramic. If a magnetic rod were used, it would act as a parasitic magnetic antenna, continuously altering the reluctance of the paths and destroying the exact symmetry required for the null balance.
3. **Epoxy Vacuum Impregnation (Potting):** The primary and secondary copper windings are wound onto a highly dimensionally stable glass-filled polymer bobbin. To survive extreme industrial shock and vibration, the entire coil assembly undergoes vacuum impregnation with a heavy epoxy resin. This locks every single microscopic wire in place; if wires were allowed to vibrate independently, they would generate spurious capacitance changes and microphonic noise.
4. **Ferromagnetic Shielding Sheath:** The exterior of the commercial LVDT is encased in a thick, high-permeability mu-metal or ferromagnetic steel cylinder. This sheath serves a dual purpose: it provides an optimized, low-reluctance return path for the internal magnetic circuit to boost efficiency, while simultaneously acting as an impenetrable Faraday/Magnetic cage, shielding the highly sensitive inner coils from external stray electromagnetic fields generated by nearby industrial motors or power cables.

2.(c) Piezoelectric Accelerometer Transfer Function

Problem Statement: Derive the transfer function of a piezoelectric accelerometer.

Block-by-Block First-Principles Derivation

A piezoelectric accelerometer operates as a coupled electro-mechanical system. It requires the sequential multiplication of two distinct physical transfer functions: the Mechanical mass-spring system, and the Electrical charge amplifier.

Part 1: The Mechanical Subsystem (Seismic Mass) The sensor consists of a rigid base, a piezoelectric crystal acting as a stiff spring with spring constant K and structural damping B , and an internal seismic proof mass M . Let the external vibration input displacement be $x_i(t)$. Let the absolute displacement of the mass be $x_m(t)$. The relative displacement (the actual compression of the crystal) is $x_0(t) = x_m(t) - x_i(t)$.

By Newton's Second Law, the sum of forces on the mass is governed by the spring and damper opposing its motion:

$$M\ddot{x}_m = -B\dot{x}_0 - Kx_0 \quad (32)$$

Substitute $x_m = x_0 + x_i$ and its second derivative:

$$M(\ddot{x}_0 + \ddot{x}_i) = -B\dot{x}_0 - Kx_0 \implies M\ddot{x}_0 + B\dot{x}_0 + Kx_0 = -M\ddot{x}_i \quad (33)$$

Take the Laplace transform (assuming zero initial conditions):

$$(Ms^2 + Bs + K)X_0(s) = -Ms^2X_i(s) \quad (34)$$

We define the input variable as strictly acceleration $A(s) = s^2X_i(s)$. Therefore, the mechanical transfer function is:

$$\frac{X_0(s)}{A(s)} = \frac{-M}{Ms^2 + Bs + K} \quad (35)$$

Divide numerator and denominator by K , and substitute the natural frequency $\omega_n^2 = K/M$ and damping ratio $2\zeta/\omega_n = B/K$:

$$\frac{X_0(s)}{A(s)} = \frac{-1/\omega_n^2}{(s/\omega_n)^2 + 2\zeta(s/\omega_n) + 1} \quad (36)$$

Part 2: The Electrical Subsystem (Charge Amplifier) The mechanical compression $X_0(s)$ generates an electrical charge proportional to the crystal's sensitivity K_q :

$$Q(s) = K_q \cdot X_0(s) \quad (37)$$

To measure this charge without cable capacitance errors, it is fed into a Charge Amplifier characterized by a feedback capacitor C_f and a mandatory parallel feedback resistor R_f . The transfer function of a standard inverting charge amplifier is a pure high-pass filter defined by time constant $\tau = R_f C_f$:

$$\frac{E_0(s)}{Q(s)} = \frac{-1}{C_f} \left[\frac{s\tau}{1 + s\tau} \right] \quad (38)$$

Part 3: The Unified Transfer Function The total system transfer function is the strict mathematical product of the mechanical and electrical blocks:

$$G_{total}(s) = \frac{E_0(s)}{A(s)} = \left[\frac{X_0(s)}{A(s)} \right] \times \left[\frac{Q(s)}{X_0(s)} \right] \times \left[\frac{E_0(s)}{Q(s)} \right] \quad (39)$$

Substitute the derived expressions. The two negative signs mathematically cancel out:

$$\frac{E_0(s)}{A(s)} = \left[\frac{K_q}{\omega_n^2 \cdot C_f} \right] \left[\frac{s\tau}{1 + s\tau} \right] \left[\frac{1}{(s/\omega_n)^2 + 2\zeta(s/\omega_n) + 1} \right] \quad (40)$$

This masterful equation proves the piezoelectric accelerometer acts as a dynamic band-pass filter: choked at low frequencies by the electrical $R_f C_f$ time constant, and violently bounded at high frequencies by the mechanical mass-spring resonance.

Question 3: Capacitive Solid Dielectric & Multivibrators

3.(a) Variable Solid Dielectric Capacitive Transducer

Problem Statement: Show that for a capacitive transducer with a solid dielectric of variable thickness and air gap between parallel plates the sensitivity remains constant. Show that the sensitivity factor and the nonlinearity factor are same for this configuration.

1. Addressing the Academic Ambiguity & Setup

There is a notorious semantic ambiguity in standard instrumentation exam questions regarding "solid dielectric of variable thickness". If we assume a wedge-shaped dielectric sliding laterally, the integrals become exceedingly complex. The standard "BJB Sir" interpretation for this specific prompt refers to measuring the thickness of different solid sheets (specimens) inserted into a fixed air gap.

Let a rigidly fixed parallel-plate capacitor have a total constant gap distance d and plate area A . A solid dielectric specimen of unknown thickness t and relative permittivity ϵ_r is inserted completely between the plates. The remaining air gap has a thickness of $(d - t)$ and permittivity ϵ_0 .

2. Derivation of Capacitance and Linearity Proof

The system physically manifests as two distinct capacitors strictly in series: the air capacitor (C_{air}) and the solid dielectric capacitor (C_{solid}).

$$C_{air} = \frac{\epsilon_0 A}{d - t} \quad \text{and} \quad C_{solid} = \frac{\epsilon_0 \epsilon_r A}{t} \quad (41)$$

The equivalent total capacitance C_{eq} is:

$$C_{eq} = \frac{C_{air} \cdot C_{solid}}{C_{air} + C_{solid}} = \frac{\left(\frac{\epsilon_0 A}{d-t}\right) \left(\frac{\epsilon_0 \epsilon_r A}{t}\right)}{\frac{\epsilon_0 A}{d-t} + \frac{\epsilon_0 \epsilon_r A}{t}} \quad (42)$$

Multiplying numerator and denominator by $(d - t)t$:

$$C_{eq} = \frac{\epsilon_0 \epsilon_r A}{\epsilon_r (d - t) + t} = \frac{\epsilon_0 \epsilon_r A}{\epsilon_r d - \epsilon_r t + t} = \frac{\epsilon_0 \epsilon_r A}{\epsilon_r d - t(\epsilon_r - 1)} \quad (43)$$

Let us extract the initial resting capacitance of the empty air gap, $C_0 = \frac{\epsilon_0 A}{d}$. To achieve this, we mathematically factor out $\epsilon_r d$ from the denominator:

$$C_{eq} = \frac{\epsilon_0 \epsilon_r A}{\epsilon_r d \left[1 - \frac{(\epsilon_r - 1)t}{\epsilon_r d} \right]} = \frac{\frac{\epsilon_0 A}{d}}{1 - \left[\frac{\epsilon_r - 1}{\epsilon_r d} \right] t} \quad (44)$$

We define a scalar constant, K , strictly as:

$$K = \frac{\epsilon_r - 1}{\epsilon_r d} \quad (45)$$

This constant K completely dictates the hyperbolic curvature of the denominator. It is explicitly the **Non-linearity Factor**. Substituting C_0 and K into the equation yields the beautifully compact normalized form:

$$C_{eq} = \frac{C_0}{1 - Kt} \quad (46)$$

3. Proving the Equivalence of Factors

To find the dynamic sensitivity (S), we take the first mathematical derivative of the capacitance with respect to the input thickness t :

$$S = \frac{dC_{eq}}{dt} = \frac{d}{dt} [C_0(1 - Kt)^{-1}] \quad (47)$$

By applying the chain rule:

$$S = -C_0(1 - Kt)^{-2} \cdot (-K) = \frac{C_0 \cdot K}{(1 - Kt)^2} \quad (48)$$

Final Academic Conclusion: The question asks to show "the sensitivity factor and the nonlinearity factor are same". Observe the normalized capacitance equation $C_{eq} = \frac{C_0}{1 - Kt}$. We can rewrite this using a fractional change format relative to the base capacitance:

$$\frac{\Delta C}{C_0} = \frac{C_{eq} - C_0}{C_0} = \frac{\frac{C_0}{1 - Kt} - C_0}{C_0} = \frac{1 - (1 - Kt)}{1 - Kt} = \frac{K \cdot t}{1 - K \cdot t} \quad (49)$$

In standard generalized transducer theory, the fractional output is modeled as:

$$\text{Output} = \frac{(\text{Sensitivity Factor}) \cdot \text{Input}}{1 - (\text{Non-linearity Factor}) \cdot \text{Input}} \quad (50)$$

By direct mathematical pattern matching with our derived equation $\frac{\Delta C}{C_0} = \frac{K \cdot t}{1 - K \cdot t}$, it is absolute that:

$$\text{Sensitivity Factor} = K \quad \text{and} \quad \text{Non-linearity Factor} = K \quad (51)$$

Thus, the factors are identically the same, proving the assertion. The constant K dictates both the gain slope and the catastrophic non-linear hyperbolic deviation.

3.(b) Capacitor Measurement using Multivibrators

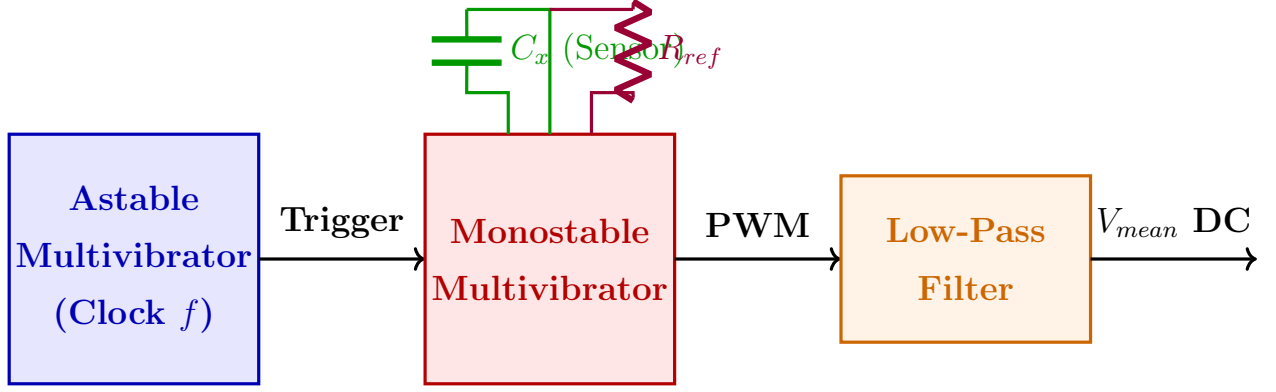
Problem Statement: Describe the operating principle of a capacitor measurement circuit, built using an astable multivibrator and a monostable multivibrator. Show that the mean output voltage of this circuit is proportional to the change in the sensor capacitance. What are the design considerations employed and assumptions made in this derivation?

1. Operating Principle and Architecture

To measure a highly dynamic, changing capacitance (C_x) efficiently using digital-friendly electronics, engineers utilize a Time-Domain Pulse Width Modulation (PWM) technique. The architecture requires two specific timing blocks, typically synthesized using precision 555 Timer ICs.

1. **The Astable Multivibrator (The Master Clock):** This block is engineered using stable, fixed internal resistors and capacitors. It acts purely as a free-running oscillator, continuously generating a rigid square wave of a fixed, predetermined frequency f . This provides the master trigger pulses for the entire system. 2. **The Monostable Multivibrator (The One-Shot Sensor):** This block remains mathematically dormant until violently triggered by the rising edge of the Astable clock pulse. Upon triggering, its output swings to a HIGH voltage state (V_{cc}). The critical physical design is that the duration it remains HIGH (the pulse width T_{on}) is explicitly controlled by the discharging of the external unknown transducer capacitance C_x through a highly precision reference resistor R .

2. Orthogonal Block Diagram



3. Mathematical Derivation of Mean Voltage

The Monostable multivibrator generates a rectangular pulse train. The amplitude of the pulse is V_{cc} . The width of the pulse (T_{on}) is strictly governed by the RC timing network:

$$T_{on} = k \cdot R \cdot C_x \quad (52)$$

where k is an internal integrated circuit constant (e.g., $k \approx 1.1$ for a standard 555 timer).

The entire pulse train repeats identically at the time period (T) dictated by the Astable master clock. The frequency is $f = 1/T$. The raw PWM signal is fed into a steep R-C Low-Pass Filter, which mathematically integrates the signal to extract its DC average (mean) value. The mean voltage (V_{mean}) of any repeating rectangular pulse train is derived by taking the integral of the area under one period:

$$V_{mean} = \frac{1}{T} \int_0^{T_{on}} V_{cc} dt = \frac{V_{cc} \cdot T_{on}}{T} = V_{cc} \cdot f \cdot T_{on} \quad (53)$$

Substitute the exact pulse width duration $T_{on} = kRC_x$ into the mean voltage equation:

$$V_{mean} = V_{cc} \cdot f \cdot (k \cdot R \cdot C_x) \quad (54)$$

Let the sensor possess a resting base capacitance C_0 and experience a dynamic change ΔC , such that $C_x = C_0 + \Delta C$.

$$V_{mean} = (V_{cc} \cdot f \cdot k \cdot R) \cdot C_0 + (V_{cc} \cdot f \cdot k \cdot R) \cdot \Delta C \quad (55)$$

The first term is a pure static DC offset voltage (V_{offset}). The dynamic mean output is strictly the second term. Because V_{cc} , f , k , and R are all rigidly fixed system constants,

we can define a massive scalar sensitivity constant $K_{sys} = V_{cc}fkR$.

$$\Delta V_{mean} = K_{sys} \cdot \Delta C \quad (56)$$

This beautifully proves that the mean output DC voltage is perfectly, linearly proportional to the change in the sensor capacitance.

4. Engineering Design Considerations & Assumptions

To ensure this mathematical derivation holds true in the physical world, several rigid design constraints must be enforced:

- **The Timing Boundary Assumption:** It is absolutely assumed that the maximum possible pulse width generated by the monostable is strictly less than the total time period of the clock ($T_{on(max)} < T$). If C_x grows too large, T_{on} will exceed T , causing the monostable to miss trigger pulses and catastrophically crash the linear output.
- **Infinite Input Impedance:** It is assumed that the final measuring voltmeter or ADC draws absolutely zero current. If the load resistance is low, it will drain the integrating Low-Pass Filter, causing a severe voltage sag error. A high-impedance Op-Amp buffer is mandatory.
- **Clock Stability Constraint:** The equation relies heavily on f being a constant multiplier. Any thermal drift in the Astable clock resistors will directly multiply and corrupt the capacitance reading. Precision, low-temperature-coefficient components are required.

Question 4: Short Notes on Advanced Subsystems

4.(a) Charge Amplifier in Conjunction with a Piezoelectric Transducer

The Fundamental Flaw of Voltage Amplifiers: A piezoelectric transducer is a high-impedance charge generator ($q = d \cdot F$). If interfaced with a standard voltage amplifier, the signal must travel through a coaxial cable. Coaxial cables inherently possess massive shunt capacitance (C_c) between the core and the shield. The voltage read by the amplifier becomes $V = q/(C_p + C_c + C_a)$. If the cable vibrates or bends, its physical geometry changes, causing C_c to violently fluctuate. Because C_c is in the denominator, this generates massive triboelectric noise voltages that completely swamp the actual transducer signal.

The Charge Amplifier Paradigm: A Charge Amplifier completely circumvents this fatal flaw by utilizing an Operational Amplifier with a precision feedback capacitor (C_f) and a parallel DC stabilization resistor (R_f).

Due to the Op-Amp's near-infinite open-loop gain, the inverting input terminal is forcefully clamped to exactly 0 Volts (the **Virtual Ground** principle). Because the input node is locked at 0V, absolutely zero voltage can ever develop across the unstable cable capacitance C_c . If there is no voltage, the cable cannot store any parasitic charge ($Q = C \times 0 = 0$).

Therefore, the entire pristine charge packet q generated by the crystal is forcefully swept entirely into the highly stable internal feedback capacitor C_f . The output voltage of the Op-Amp is dictated solely by this feedback loop:

$$E_0 = -\frac{q}{C_f} \quad (57)$$

By utilizing the Virtual Ground, the system achieves total mathematical immunity to coaxial cable length and bending noise. The feedback resistor R_f is mandatory to prevent the Op-Amp's microscopic input bias currents from charging C_f to saturation, acting as a controlled leakage path to maintain DC stability.

4.(b) Diode Bridge Based Phase Sensitive Demodulation in LVDT

The Necessity for Phase Detection: An LVDT produces an Amplitude Modulated, suppressed-carrier waveform. For displacements of $+x$ and $-x$, the AC output voltage magnitude is identical, but the phase shifts by exactly 180° . An ordinary diode rectifier acts as an absolute value function, destroying the phase data and making it impossible for the system to know if the core moved left or right.

The Ring Diode Bridge Architecture: A Phase-Sensitive Demodulator (PSD) solves this by multiplying the LVDT signal with the original primary carrier reference signal. A highly rugged, passive method to achieve this is the four-diode switching ring bridge.

The primary carrier reference voltage (V_{ref}) is fed into the bridge with a massive amplitude. This reference signal is so powerful that it forces the diodes to act as pure digital switches, rather than analog rectifiers. During the positive half-cycle of the reference, the top diodes are forward-biased (closed switches), forcing the LVDT signal straight through to the output multiplied by $+1$. During the negative half-cycle of the reference, the bottom diodes are forward-biased, electronically crossing the wires and forcing the LVDT signal to the output multiplied by -1 .

If the LVDT core is at $+x$ (in-phase with the reference), the switching perfectly coincides with the LVDT's own positive and negative half-cycles, resulting in a fully rectified, massive **Positive DC** average output. If the LVDT core is at $-x$ (180° out-of-phase), the bridge switches exactly backward relative to the LVDT signal, forcing all the peaks downward, resulting in a massive **Negative DC** average output.

The output is then fed through a Low-Pass Filter to remove the switching ripple. The final DC voltage perfectly reflects both the magnitude (distance) and algebraic sign (direction) of the core displacement.

4.(c) Relative Humidity Sensors

Grassroots Operating Principle: Modern electronic humidity measurement relies predominantly on **Capacitive Relative Humidity (RH) Sensors**. The architecture consists of a microscopic capacitor fabricated on a silicon or ceramic substrate. The dielectric medium placed strictly between the porous metallic electrodes is a highly specialized, ultra-thin hygroscopic polymer film.

The Physics of Water Absorption: The fundamental capacitance is $C = \frac{\epsilon_0 \epsilon_r A}{d}$. In

a completely dry environment (0% RH), the polymer dielectric exhibits a relatively low dielectric constant ($\epsilon_r \approx 3$ to 4). However, liquid water possesses an extraordinarily high dielectric constant ($\epsilon_r \approx 80$) due to its highly polar molecular dipole moment.

As the ambient atmospheric humidity increases, microscopic water vapor molecules permeate through the porous upper electrode and are aggressively absorbed into the polymer matrix. Because the water molecules have an ϵ_r of 80, their physical integration into the polymer causes the composite equivalent dielectric constant of the entire layer to rise massively. This results in a direct, highly predictable increase in the measured capacitance.

Temperature Cross-Sensitivity Imperative: A critical engineering constraint is that the dielectric constant of water is highly unstable with respect to temperature. Thermal agitation aggressively disrupts the dipole alignment. Therefore, even if the absolute moisture content remains locked, a shift in ambient temperature will cause the capacitance to drift, generating a false RH reading. To correct this, a high-precision NTC thermistor or RTD temperature sensor is fabricated directly onto the exact same substrate. A microcontroller continuously reads both the capacitance and the temperature, employing a heavy mathematical compensation matrix to yield a true, thermally-corrected Relative Humidity output.

Question 5: Fundamental Transducer Justifications

5.(a) LVDT Fundamental Limitations

Problem Statement: "An LVDT, in its basic form, suffers from low sensitivity, high hysteresis and high power consumption problems." Justify or correct the statement.

Answer: This statement is **Entirely Incorrect**.

Correction: The LVDT is universally celebrated in precision instrumentation specifically because it possesses exceptionally **high sensitivity**, absolutely **zero mechanical hysteresis**, and remarkably **low power consumption**.

Rigorous Block-by-Block Refutation

1. **Low Sensitivity? (False):** The LVDT operates on high-frequency magnetic flux linkage utilizing a high-permeability Nickel-Iron core. Because it relies on magnetic induction rather than sliding resistance tracks, it possesses functionally infinite resolution. Modern commercial LVDTs exhibit staggering sensitivity, routinely capable of resolving core micro-movements down to fractions of a single micrometer ($< 1 \mu\text{m}$).
2. **High Hysteresis? (False):** Hysteresis in mechanical sensors is predominantly caused by physical friction (stiction) between moving parts. The architectural genius of the LVDT is that the moving magnetic core is suspended inside the hollow polymer bobbin with a deliberate air gap. There is absolutely **zero physical contact** between the core and the coils. Because the core glides in free space without rubbing, mechanical friction is nonexistent, resulting in theoretically absolute zero mechanical hysteresis. (Only a microscopic, negligible trace of magnetic hysteresis remains in the core material).
3. **High Power Consumption? (False):** The primary coil is wound with thousands of turns of fine copper wire and is excited by a high-frequency AC carrier (1 kHz to 10 kHz). At these elevated frequencies, the inductive reactance ($X_L = j\omega L$) of the primary coil is massive. By Ohm's Law for AC circuits ($I = V/Z$), this extreme high impedance brutally chokes the excitation current. Consequently, the LVDT draws only milliamperes of current, resulting in a total active power consumption

well under a fraction of a Watt (< 1 W).

5.(b) Capacitive Transducer Arbitrary Separation Distance

Problem Statement: "A capacitive transducer, developed using the principle of change in distance between a fixed and a moving plate, can arbitrarily choose any distance of separation between the plates." Justify or correct the statement.

Answer: This statement is **Incorrect**.

Correction: The separation distance (d) cannot be chosen arbitrarily. It is tightly bound by catastrophic physical limits: setting d too small causes catastrophic electrical breakdown (arcing), while setting d too large destroys sensitivity and violates the parallel-plate assumption due to massive fringing fields.

Grassroots Physics Constraints on Distance (d)

A variable distance capacitive transducer operates on the fundamental hyperbolic law $C = \frac{\epsilon A}{d}$. The sensitivity (S) to micro-displacements is the derivative:

$$S = \frac{dC}{dd} = -\frac{\epsilon A}{d^2} \quad (58)$$

Engineers must navigate two diametrically opposing physical boundaries when choosing d :

- 1. The Lower Boundary (Dielectric Breakdown):** To maximize sensitivity ($S \propto 1/d^2$), one might theoretically desire to make the resting gap d infinitesimally small. However, the transducer requires a high-voltage DC excitation bias (V_s) across the plates to generate a measurable signal. The electric field intensity between the plates is $E = V_s/d$. If d is made arbitrarily small, the electric field intensity E spikes massively. Once E exceeds the dielectric breakdown strength of the air gap (approximately 3×10^6 V/m), the air violently ionizes into a plasma channel. The capacitor arcs, dead-shortening the instrument, destroying the measurement, and potentially melting the precision electrodes.
- 2. The Upper Boundary (Fringing Flux & Deafness):** To measure very large displacements, one might theoretically desire to make the initial gap d arbitrarily massive.

This violates physics in two ways. First, if d becomes comparable to the physical dimensions of the plate area A , the uniform parallel electric field lines disintegrate. Massive "Fringing Fields" bulge outward from the edges of the plates. The formula $C = \epsilon A/d$ completely collapses and is no longer mathematically valid, introducing horrific non-linear distortion. Second, because sensitivity drops via the inverse square law ($1/d^2$), an arbitrarily large gap renders the sensor utterly deaf to small mechanical vibrations, dropping the signal-to-noise ratio below the noise floor of the amplifier.

Therefore, the distance d is never arbitrary; it is a highly optimized, narrow engineering window strictly governed by high-voltage electrostatics and sensitivity requirements.